

A spectral bound on relaxation of the trajectory-averaged circuit

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We obtain the two-point relaxation rate directly from an ordered product of single-particle projectors. No trajectory sampling or explicit time evolution is required. This rate bounds the full channel's relaxation rate from above.

We ask how quickly the circuit forgets its initial two-point function after discarding the measurement record. We use the manuscript convention $C_{ab} = \text{Tr}(\hat{\rho}\hat{c}_a^\dagger\hat{c}_b)$; the canonical CPU engine stores the centered matrix $G = 2C - \mathbf{1}$. For a normalized overcomplete Wannier (OW) mode j , let P_j be its projector in the stored correlation ordering and $Q_j = \mathbf{1} - P_j$. Perfect feedback fills a lower-band mode ($s_j = 1$) or depletes an upper-band mode ($s_j = 0$). Gain or loss contributes $s_j P_j - \{P_j, \bar{C}\}/2$ to the covariance increment, while number dephasing contributes $-\{P_j, \bar{C}\}/2 + P_j \bar{C} P_j$. The elementary channel $\mathcal{E}_j = \text{Id} + \mathcal{L}_j$ therefore gives

$$\bar{C}' = \bar{C} - \{P_j, \bar{C}\} + P_j \bar{C} P_j + s_j P_j = Q_j \bar{C} Q_j + s_j P_j. \quad (1)$$

This is an exact finite reset, not a time-step approximation. Closure of this equation does not require the averaged state to be Gaussian.

One cycle is the composition $\mathcal{E}_{j_M} \cdots \mathcal{E}_{j_1}$, with the rightmost operation acting first. At cycle boundaries,

$$\bar{C}_{n+1} = A \bar{C}_n A^\dagger + B, \quad A = Q_{j_M} \cdots Q_{j_1}, \quad \delta C_{n+1} = A \delta C_n A^\dagger, \quad (2)$$

where B is independent of the input and $\delta C = \bar{C} - C_{\text{ss}}$. Column vectorization yields $T = A^* \otimes A$. If a_p are the eigenvalues of A , the covariance multipliers are $a_p a_q^*$. Hence, provided $0 < \rho(A) < 1$,

$$\rho(A) = \max_p |a_p|, \quad \Delta_C = -\log \rho(T) = -2 \log \rho(A) = \min_{p,q} [-\log |a_p a_q^*|]. \quad (3)$$

The largest survival factor gives the smallest decay rate, measured per cycle. This is not the occupation-spectrum gap around $1/2$, nor the largest singular value of A . Non-normality can affect transients; Jordan blocks add polynomial prefactors without changing the exponential rate.

The adjoint channel preserves $\mathcal{V}_2 = \text{span}\{\hat{\mathbf{1}}, \hat{c}_a^\dagger \hat{c}_b\}$, and its restriction has diagonal blocks 1 and $T^\top$. Spectral inclusion then implies, for a mixing full channel in the same physical symmetry sector,

$$r_{\text{full}} \geq \rho(T), \quad \Delta_{\text{full}} = -\log r_{\text{full}} \leq \Delta_C. \quad (4)$$

Mixing means a unique stationary state and no other unit-modulus eigenvalues in that sector; r_{full} excludes the stationary eigenvalue. A closing upper bound forces slow relaxation, but a finite bound does not establish a many-body gap: higher-order observables may relax more slowly.

We evaluate $N_x = 20$, $N_y = 20, 24, 28, 32, 36, 40, 44, 50$, and $\alpha_1 = 1, 3$, with $\alpha_2 = 30$, $n_{\text{shell}} = 1$, hard-wall support truncation at the inclusive slab $x = 5, \dots, 15$, and all slabs active. Both directions are periodic, the twist is zero, and the trial orbitals are the X pair. The fixed raster- y schedule advances y at fixed x , with within-cell order $A+, A-, B+, B-$. We construct A by complex128 local rank-one updates and compute its full non-Hermitian spectrum. There is no sampled ensemble, chosen initial covariance, temporal average, or fitted decay law. Across these eight circumferences, we find $\Delta_C \in [1.829759, 1.904385]$ for $\alpha_1 = 1$ and $\Delta_C \in [3.183716, 3.250052]$ for $\alpha_1 = 3$, in inverse cycles. All 16 cases have resolved positive covariance rates; this finite-size observation is not a proof of a thermodynamic or many-body gap. The largest independent dominant-eigenpair residual is 5.45×10^{-15} .